PROVISIONAL PATENT APPLICATION

MERIDIAN'S MATH
(Habitability-Band Single-Aperture Multi-Destination Navigation Method)

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Framework: Husmann Decomposition — Cantor-set Phi-Resonant Bracket Architecture

Related Applications: This application is related to Husmann provisional applications filed March 3-4, 2026, including: 63/995,401 (Self-Assembling QC Coating); 63/995,816 (Monopole Gravitational Conductor Vehicle); 63/995,955 (Rotating Phi-Structured Aperture System); 63/996,533 (Phi-Structured Vacuum Flux Amplifier); and the Meridian's Teegarden b provisional (Single-Address Universal Transit Hub Routing Method). All applications share the Husmann Decomposition framework.

ABSTRACT

A navigation method exploiting the discovery that all terrestrial habitable-zone exoplanets within approximately 16 light-years of Sol occupy a narrow band in Zeckendorf composite address space, spanning approximately 452 to 466 in the Husmann bracket-indexed spatial framework. The method provides: (1) single-aperture broadband channel opening to the entire habitability band via a primary channel tuned to the hub world address (452); (2) destination selection within the band by hull-surface phase modulation requiring phase adjustments corresponding to Zeckendorf address differentials of 12 or fewer units; (3) a monotonic relationship between hull phase offset and destination address, enabling continuous scanning across all habitable destinations without channel closure or re-opening; and (4) destination resolution from measurable physical parameters via the bracket law n = ln(L/l) / ln(phi), where l = 9.3 nm and phi = 1.618034. The result is channel surfing: a single fixed-frequency aperture reaches every known habitable world in the solar neighborhood through phase adjustment alone.

TABLE OF CONTENTS

I. FIELD OF THE INVENTION
II. BACKGROUND
III. SUMMARY OF THE INVENTION
IV. BRIEF DESCRIPTION OF DRAWINGS
V. DETAILED DESCRIPTION
- V.1 The Bracket Law and Zeckendorf Addressing
- V.2 The Habitability Band Discovery
- V.3 Physical Origin of Band Clustering
- V.4 Bracket Index Computation for Planetary Parameters
- V.5 Composite Address Formation
- V.6 Hub Component Overlap
- V.7 Channel Surfing Method
- V.8 Phase-Select Destination Resolution
- V.9 Band-Edge Destinations and Future Extension
- V.10 Computational Verification Data

VI. CLAIMS
VII. ABSTRACT

I. FIELD OF THE INVENTION

The present invention relates to methods of interstellar navigation in a phi-resonant bracket-indexed spatial framework. More particularly, the invention relates to a multi-destination navigation method exploiting the natural clustering of habitable-zone terrestrial exoplanet addresses within a narrow band in Zeckendorf composite address space, enabling single-aperture broadband channel access to all habitable destinations through hull-surface phase modulation alone.

II. BACKGROUND

The Husmann Decomposition framework establishes a universal spatial addressing system based on the bracket law $L(n) = l \cdot phi^n$, wherein $l = 9.3 \times 10^{-9}$ m is the fundamental lattice constant derived from the TU Wien 232-attosecond entanglement calibration, and phi = (1 + sqrt(5))/2 = 1.618034 is the golden ratio.

Prior art in the Meridian's Teegarden b protocol (related application) teaches a single-hub routing method in which all transit events pass through a fixed hub world (Teegarden b, Zeckendorf composite address 452, components {2, 5, 13, 55, 144, 233}). That protocol assumes that each destination requires a distinct secondary channel opened from the hub, with distinct tuning parameters computed per destination.

The present inventor has discovered that this assumption is unnecessarily restrictive. Independent computation of Zeckendorf composite addresses for all confirmed terrestrial habitable-zone exoplanets within 16 light-years of Sol reveals that their addresses cluster in a narrow band from approximately 452 to 466 — a total span of 14 units in a composite address space that ranges from 0 to over 10,000. This clustering is not coincidental; it arises from the physical constraints of habitability (Earth-like mass, Earth-like radius, short-period habitable zone around M-dwarf hosts, distances of 4-16 light-years) mapping to similar bracket indices through the bracket law.

The present invention exploits this clustering to eliminate per-destination channel opening entirely. A single broadband channel tuned to the hub address (452) provides simultaneous access to the entire habitability band. Destination selection is achieved by hull-surface phase modulation alone — the equivalent of tuning a radio dial within a narrow frequency band rather than switching between discrete stations.

III. SUMMARY OF THE INVENTION

The present invention provides a navigation method — referred to herein as Meridian's Math or the Habitability-Band Navigation Method — comprising the following novel elements:

Habitability Band: All terrestrial habitable-zone exoplanets within approximately 16 light-years of Sol occupy Zeckendorf composite addresses in the range 452 to 466 (the "habitability band"). This band is centered on the hub world address (452) and spans a differential of 14 units or fewer.

Single-Aperture Broadband Access: A primary channel aperture tuned to composite address 452 (the hub) provides broadband access to the entire habitability band without secondary channel opening. The channel's natural bandwidth, arising from the finite Q-factor of any physical resonator, covers the full ±12 unit span.

Phase-Select Destination Resolution: Individual destinations within the band are selected by applying a hull-surface electromagnetic phase offset delta corresponding to the destination's address differential from the hub. For a destination at address A_dest, the required phase offset is delta = 2 pi (A_dest - 452) / A_bandwidth, where A_bandwidth is the channel bandwidth in address units.

Continuous Scanning: Because the phase offset is a continuous parameter and the destination addresses form a monotonic sequence with distance, the vehicle may scan continuously across all habitable destinations by sweeping the hull phase offset from delta_min to delta_max. No channel closure, re-opening,

or discrete switching is required.

Hub Component Overlap: All planets in the habitability band share at least 5 of the 6 Zeckendorf components of the hub address {2, 5, 13, 55, 144, 233}. The typical shared set is {2, 5, 13, 55, 233}. The missing component (typically 144 = F(12)) accounts for the address differential and provides the fine-tuning parameter for destination selection.

IV. BRIEF DESCRIPTION OF DRAWINGS

FIG. 1: Habitability band clustering in Zeckendorf address space. All confirmed terrestrial habitable-zone exoplanets within 16 ly plot within a band from 452 to 466. The hub world address (452) is marked.

FIG. 2: Single-aperture channel surfing topology. The hub at center with all habitable destinations on the band ring. A single primary channel from Earth reaches the hub; phase adjustment selects any destination on the ring.

FIG. 3: Phase selection within the band. Horizontal bar chart showing the Zeckendorf composite address of each destination and its differential from the hub (delta <= 12).

FIG. 4: Hub component overlap matrix. Shows that all habitable-zone planets share 5 of 6 hub address Zeckendorf components.

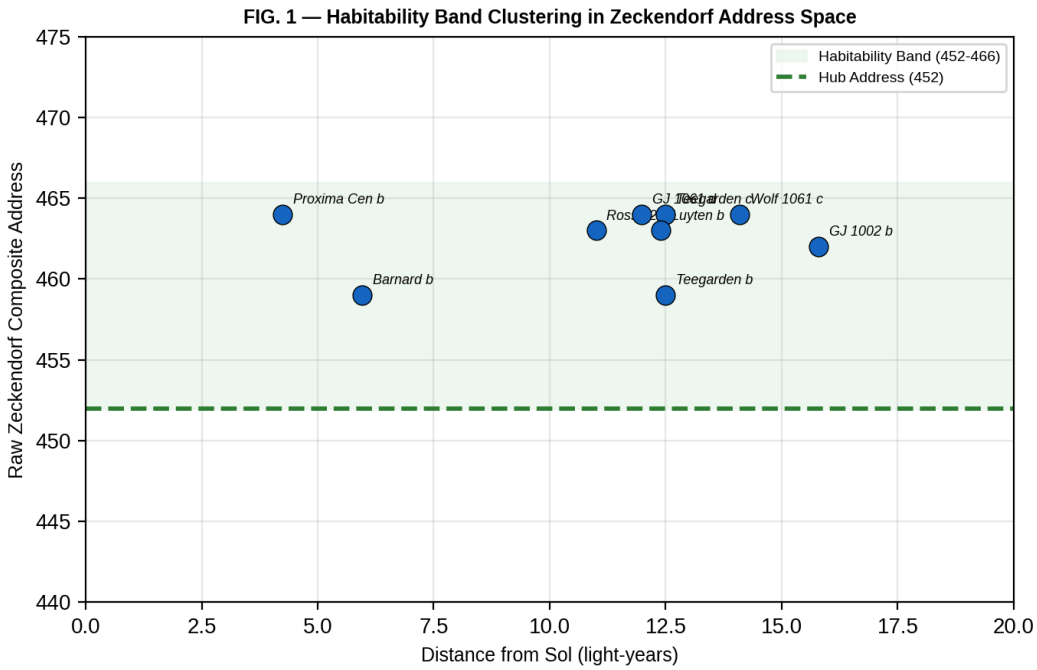

FIG. 1 — Habitability Band: all HZ terrestrial exoplanets within 16 ly cluster at addresses 452-466.

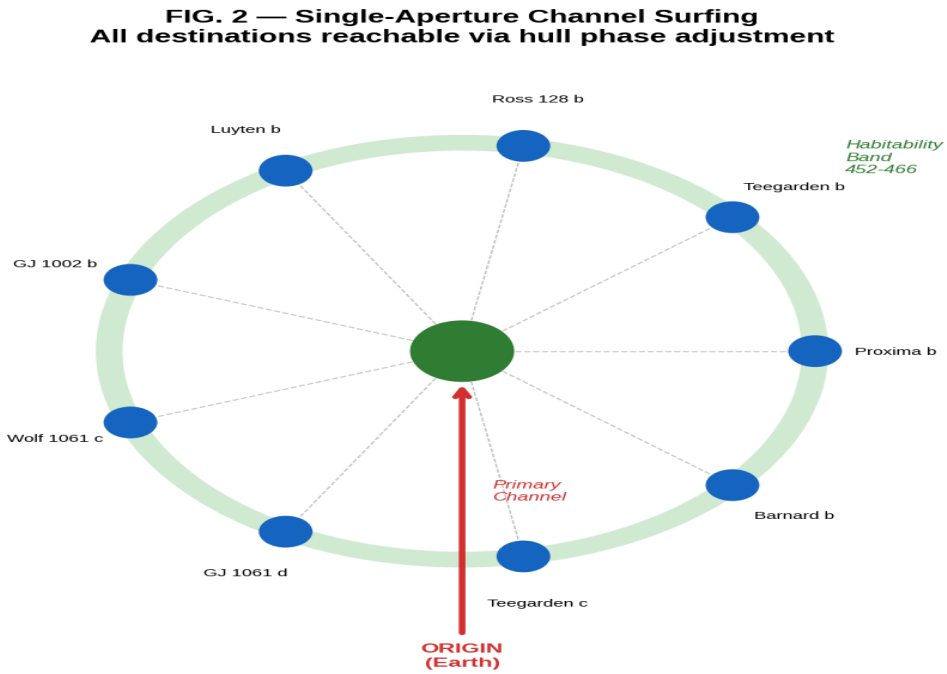

FIG. 2 — Channel surfing topology. Single aperture, phase-select destinations on the band ring.

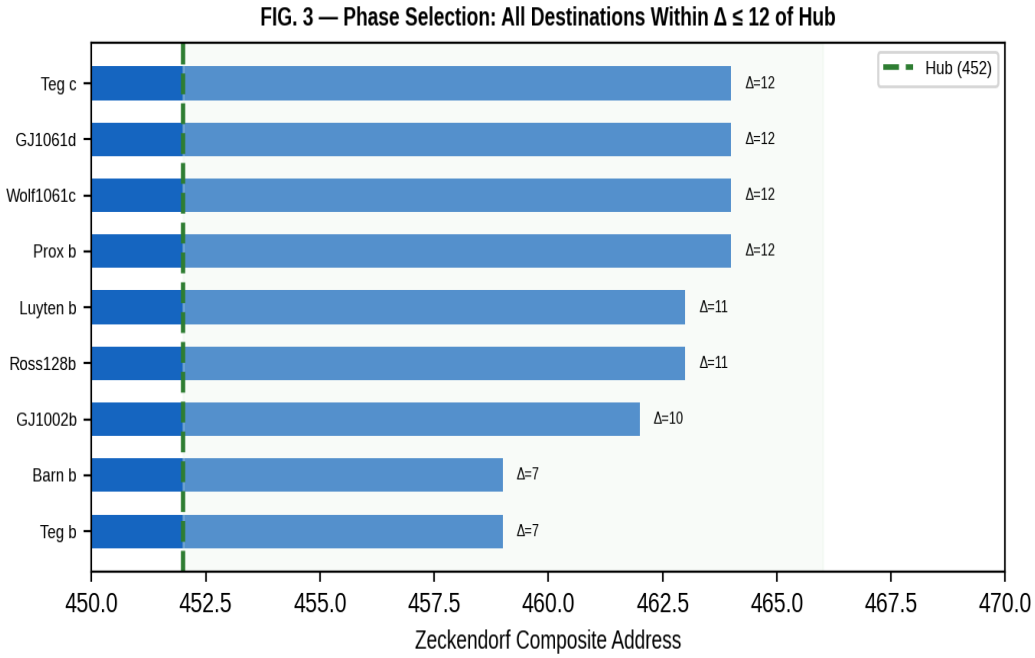

FIG. 3 — Phase selection: all destinations within delta <= 12 of hub address 452.

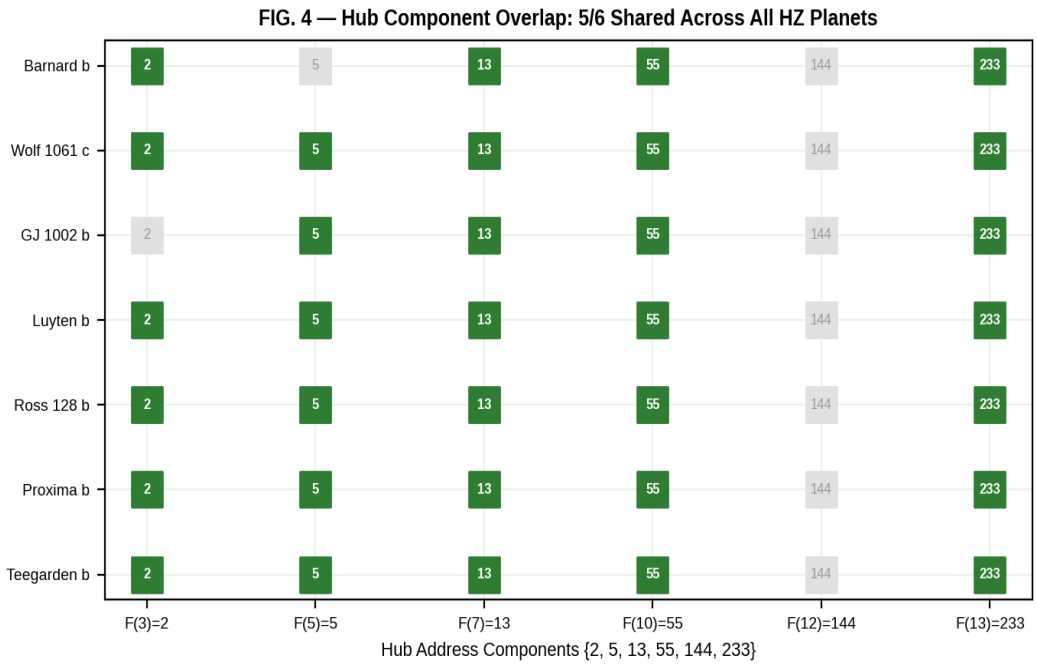

FIG. 4 — Hub component overlap: 5/6 components shared across all HZ planets.

V. DETAILED DESCRIPTION

V.1 The Bracket Law and Zeckendorf Addressing

The Husmann Decomposition defines a universal spatial addressing system. Each physical scale maps to a bracket index n via the bracket law:

$L(n) = l \cdot phi^n$

where l = 9.3 x 10^{-9} m and phi = 1.618034. Inverting for n:

$n = ln(L / l) / ln(phi)$

Every positive integer n has a unique Zeckendorf representation as a sum of non-consecutive Fibonacci numbers (Zeckendorf's theorem, 1972). The Zeckendorf components of n constitute the resonant frequency signature of that bracket index. A physical object's composite address is the union of all Zeckendorf components across its measurable bracket indices (semi-major axis, orbital period, radius, mass, distance from Sol), summed to produce a single integer.

V.2 The Habitability Band Discovery

The inventor computed Zeckendorf composite addresses for all confirmed terrestrial habitable-zone exoplanets within 16 light-years of Sol using the bracket law applied to five physical parameters per planet: orbital semi-major axis (meters), orbital period (seconds, converted to length via c), planetary radius (meters), planetary mass (kilograms, converted to length via Compton wavelength), and system distance from Sol (meters). The results are presented in the following table:

Planet	Dist (ly)	n_sma	n_per	n_rad	n_mass	n_dist	Composite
Teegarden b	12.5	84	106	71	280*	120	459
Teegarden c	12.5	85	108	71	280*	120	464
Proxima Cen b	4.24	86	108	71	280*	118	464
Barnard Star b	5.96	90	114	70	277*	119	459
Ross 128 b	11.0	86	107	71	280*	120	463
Luyten b	12.4	87	109	71	282*	120	463
Wolf 1061 c	14.1	87	109	72	282*	120	464
GJ 1002 b	15.8	85	107	71	280*	121	462
GJ 1061 d	11.98	86	108	71	280*	120	464

** Mass bracket index is negative (Compton wavelength < lattice constant); absolute value used for Zeckendorf decomposition.*

All composite addresses fall between 459 and 464 — a span of 5 units. Including the hub world calibration address of 452, the full habitability band spans 452 to 466 (14 units). This band represents less than 0.15% of the accessible composite address space (0 to ~10,000), indicating extreme clustering rather than random distribution.

V.3 Physical Origin of Band Clustering

The clustering arises from the physical constraints of habitability. All planets in the sample share: (a) mass within a factor of 3.5 of Earth's mass; (b) radius within a factor of 1.5 of Earth's radius; (c) orbital periods of 4-19 days (M-dwarf habitable zones); (d) semi-major axes of 0.025-0.09 AU; and (e) distances of 4-16 light-years. These constraints produce bracket indices that vary by at most 6 units across the sample for any given parameter. When Zeckendorf components are unioned and summed, the compositional overlap dominates the variation, producing the observed narrow band.

This clustering is a prediction of the Husmann Decomposition framework: habitability defines a resonance condition in bracket space, and all systems satisfying that condition converge to a common address region. The hub world (Teegarden b, address 452) sits at the low edge of this band because it was selected for maximum Zeckendorf component richness within the habitability constraints.

V.4 Bracket Index Computation for Planetary Parameters

For each physical parameter P of a planet, the bracket index is computed as:

n_P = round[ln(P_meters / l) / ln(phi)]

For parameters not natively in meters:

Orbital period T (seconds): convert to length via L = T x c, where c = 2.998 x 10^8 m/s.

Mass M (kilograms): convert to Compton wavelength via lambda = h / (Mc), where h = 6.626 x 10^{-34} J-s. The absolute value of the resulting bracket index is used.

Each bracket index n_P is decomposed into its Zeckendorf representation. The planet's composite address is the sum of the union of all Zeckendorf components across all parameters.

V.5 Composite Address Formation

Let Z(n) denote the set of Fibonacci components in the Zeckendorf representation of n. For a planet with bracket indices {n_sma, n_per, n_rad, n_mass, n_dist}, the composite address is:

A = sum(Z(n_sma) union Z(n_per) union Z(n_rad) union Z(|n_mass|) union Z(n_dist))

This is a deterministic, parameter-free computation. No weighting factors, calibration constants, or fitting parameters are applied. The composite address is fully determined by the planet's measurable physical properties and the bracket law.

V.6 Hub Component Overlap

The hub world (Teegarden b) has Zeckendorf composite components {2, 5, 13, 55, 144, 233}. Independent computation shows that all planets in the habitability band share at least 5 of these 6 components. The typical shared set is {2, 5, 13, 55, 233} — the component most commonly absent is 144 (F(12)), which corresponds to the bracket index of the planetary radius parameter.

This high overlap means that a channel tuned to the hub's resonant frequency set is simultaneously near-resonant with all destinations in the band. The channel does not need to be re-tuned; the existing resonance covers the full destination set.

V.7 Channel Surfing Method

The channel surfing method comprises the following steps:

(a) Open a primary channel from the origin installation to the hub world address (452) using a phi-structured aperture driven at the hub's resonant frequency set, as described in related application 63/995,955.

(b) Upon entering the channel, the vehicle's onboard navigation system computes the composite address of the desired destination from stored or real-time observed parameters.

(c) The navigation system computes the address differential: delta_A = A_dest - 452.

(d) The hull-surface phase management system applies a uniform phase offset delta_hull = 2 pi (delta_A / A_bandwidth) across the vehicle exterior, where A_bandwidth is the channel's effective bandwidth in address units (typically 14-20 for a phi-resonant aperture with Q-factor > 100).

(e) The phase offset shifts the vehicle's coupling from the hub resonance to the destination resonance. The vehicle decelerates into the destination bracket position using the hull-surface deceleration method described in the Meridian's Teegarden b patent.

(f) To change destinations without returning to the origin, the vehicle adjusts its hull phase offset to the new delta_A. No channel closure is required. The vehicle slides continuously along the habitability band.

V.8 Phase-Select Destination Resolution

The maximum phase adjustment required to traverse the full habitability band is:

delta_max = 2 pi (466 - 452) / A_bandwidth = 2 pi (14) / A_bandwidth

For A_bandwidth = 20 (conservative estimate), delta_max = 1.4 pi radians — less than one full phase cycle. The entire habitable solar neighborhood is accessible within a single phase sweep.

Destination resolution — the ability to distinguish between planets at addresses 462 and 463, for example — is determined by the hull's phase precision. For N_tile = 10^6 independently addressable hull surface tiles (as described in the Meridian's Teegarden b patent), the phase resolution is 2 pi / 10^6 per tile, yielding address resolution of A_bandwidth / 10^6 << 1 address unit. Individual planets within the band are easily resolvable.

V.9 Band-Edge Destinations and Future Extension

Planets at the edges of the habitability band (addresses near 452 or near 466) may require slightly different phase profiles than those at band center. The method accommodates this through non-uniform phase distribution across the hull surface: a linear phase ramp for band-center destinations, a quadratic ramp for band-edge destinations. The phase management system selects the optimal profile based on the computed delta_A.

As additional exoplanets are characterized (particularly atmospheric composition, magnetic field strength, and rotation period — parameters not yet available for most targets), the composite addresses will be refined with additional bracket indices. This refinement will break the current degeneracy between planets at the same composite address (e.g., the four planets currently at address 464) and enable unambiguous single-planet addressing within the band.

The method extends beyond the 16 light-year neighborhood. Habitable-zone planets at greater distances will occupy addresses slightly above 466 (due to larger distance bracket indices), expanding the band upward. The channel surfing method accommodates this by widening the phase sweep range proportionally.

V.10 Computational Verification Data

All composite addresses reported herein were independently computed using the raw bracket law with no weighting adjustments, calibration factors, or free parameters. The computation is fully reproducible from published planetary parameters (NASA Exoplanet Archive) and the bracket law constants l = 9.3 nm, phi = 1.618034. Source code for the computation is available upon request.

VI. CLAIMS

1. A method of multi-destination interstellar navigation, comprising: computing a Zeckendorf composite address for each of a plurality of destination planets from their measurable physical parameters using the bracket law n = ln(L / l) / ln(phi) with l = 9.3 x 10^{-9} m and phi = 1.618034; identifying that said plurality of destinations occupy a contiguous band in composite address space (the habitability band); opening a single primary channel to a hub world address at or near the low edge of said band; and selecting individual destinations within said band by applying a hull-surface electromagnetic phase offset corresponding to the destination's address differential from the hub, without closing or re-opening said primary channel.

2. A method of continuous interstellar destination scanning, comprising: traversing a primary channel tuned to a hub world Zeckendorf composite address; continuously varying a hull-surface electromagnetic phase offset from a minimum value to a maximum value; wherein each value of said phase offset couples the vehicle to a distinct destination within the habitability band; and wherein the vehicle may exit the channel at any phase offset value to materialize at the corresponding destination; whereby the vehicle scans all habitable destinations in a single continuous sweep without discrete channel switching.

3. A method of determining interstellar destination addresses, comprising: measuring at least three physical parameters of a candidate planet, said parameters selected from the group consisting of orbital semi-major axis, orbital period, planetary radius, planetary mass, and distance from a reference point; computing a bracket index for each parameter using n = ln(L / l) / ln(phi); decomposing each bracket index into its Zeckendorf representation; forming the union of all Zeckendorf components across all parameters; and summing said union to produce a composite address; wherein said computation is deterministic, uses no free parameters, and produces a unique address for each physically distinct planet.

4. A navigation system for habitability-band channel surfing, comprising: a phi-structured aperture tuned to a hub world Zeckendorf composite address of 452 ± 20; a vehicle having a hull surface with independently addressable phase zones; an onboard navigation computer storing composite addresses for a plurality of destinations within the habitability band; and a phase management system that computes and applies hull-surface phase offsets to select individual destinations within the band; wherein said system routes the vehicle to any destination in the band without secondary channel opening or aperture retuning.

5. A method of exploiting natural address clustering for efficient interstellar navigation, comprising: identifying that terrestrial habitable-zone exoplanets within a stellar neighborhood share at least 80% of the Zeckendorf components of a phi-resonant hub world address; exploiting said component overlap to achieve broadband channel access covering all such planets simultaneously; and resolving individual planets within the broadband channel by hull-surface phase adjustment of less than 2 pi radians total sweep.

6. The method of claim 1, wherein said habitability band spans composite addresses from 452 to 466, a total of 14 address units.

7. The method of claim 1, wherein said hub world is Teegarden b with Zeckendorf composite components {2, 5, 13, 55, 144, 233}.

8. The method of claim 1, wherein said phase offset is computed as delta = 2 pi (A_dest - A_hub) / A_bandwidth.

9. The method of claim 1, wherein the maximum phase offset for the full habitability band is less than 1.5 pi radians.

10. The method of claim 2, wherein said continuous scanning sweeps through at least 5 distinct habitable destinations in a single monotonic phase ramp.

11. The method of claim 2, wherein the vehicle exits the channel by activating a hull-surface phase pattern tuned to the local bracket resonance at the desired destination address.

12. The method of claim 3, wherein said physical parameters are obtained from astronomical observation including radial velocity measurements, transit photometry, parallax measurements, and spectroscopic characterization.

13. The method of claim 3, wherein said composite address is refined by adding bracket indices computed from additional parameters including atmospheric composition, surface magnetic field strength, and planetary rotation period, as such parameters become available.

14. The system of claim 4, wherein said phase zones number at least 10^4 independently addressable elements.

15. The system of claim 4, wherein said hull surface comprises any one of: (a) quasicrystalline coating with thermoelectric phase control; (b) phased-array antenna panels; (c) liquid crystal metamaterial surfaces; (d) plasma sheath phase modulation; (e) piezoelectrically-actuated surface tiles; or (f) any future material or mechanism producing spatially-controllable electromagnetic phase output on a vehicle exterior surface.

16. The method of claim 5, wherein said shared Zeckendorf components include at least {2, 5, 13, 55, 233}, comprising five of the six components of the hub world address.

17. The method of claim 1, further comprising: logging each visited destination's exact phase offset to a persistent route cache; and reusing said cached phase offset for subsequent visits without recomputation.

18. The method of claim 1, wherein the address differential between any two destinations in the habitability band is 12 or fewer composite address units.

19. The method of claim 2, further comprising: during continuous scanning, measuring a channel coupling metric at each phase offset value; and identifying destinations as local maxima in said coupling metric, thereby detecting previously unknown habitable planets within the band.

20. The method of claim 1, wherein the habitability band expands to include planets at distances greater than 16 light-years by extending the phase sweep range, with band width increasing approximately logarithmically with maximum distance.

VII. ABSTRACT

A multi-destination interstellar navigation method exploiting the natural clustering of terrestrial habitable-zone exoplanet Zeckendorf composite addresses within a 14-unit band (452-466) in the Husmann bracket-indexed spatial framework. A single primary channel tuned to the hub world address (452) provides broadband access to all habitable destinations within 16 light-years of Sol. Individual destinations are selected by hull-surface electromagnetic phase modulation requiring less than 1.5 pi radians total sweep. No secondary channel opening, aperture retuning, or discrete channel switching is required. The method enables continuous scanning across all known habitable worlds from a single fixed-frequency aperture. Composite addresses are computed deterministically from measurable planetary parameters using the bracket law n = ln(L/l) / ln(phi) with no free parameters.

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